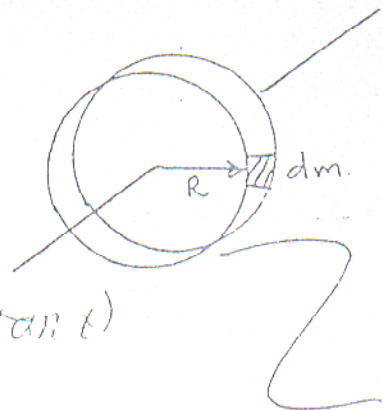


- (c) 'n Dun ring met massa m en straal R roteer om 'n as deur sy middelpunt. Toon aan dat die traagheidsmoment van die hele ring is $I = mR^2$.

A thin ring has a mass m and radius R and it rotates about an axis through its center. Show that the moment of inertia of the whole ring is $I = mR^2$.

$$\begin{aligned}
 I &= \int R^2 dm \\
 &= R^2 \int dm \quad (\text{want } R \text{ bykonstant}) \\
 &= R^2 m \\
 &= mR^2
 \end{aligned}$$

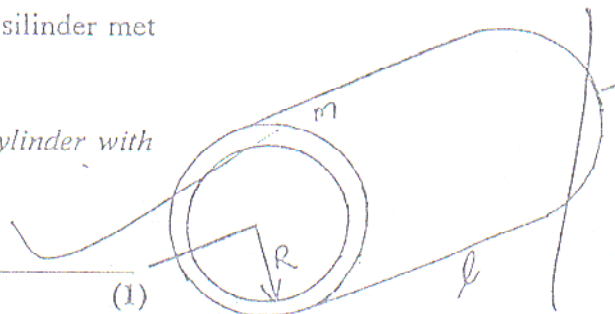


(2)

- (d) Wat is die traagheidsmoment van 'n dun silinder met straal R , massa m en lengte ℓ ?

What is the moment of inertia of a thin cylinder with radius R , mass m and length ℓ ?

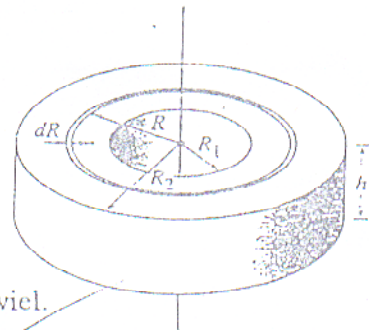
$$I = mR^2$$



(1)

- (e) 'n Sloopwiel het 'n gat in die middel met 'n binne radius R_1 , 'n buiteradius R_2 en 'n massa M .

A grindstone with a hole in the center has an inner radius R_1 , an outer radius R_2 and a mass M .



- (i) Skryf 'n uitdrukking neer vir die volume V van die wiel.

Give an expression for the volume V of the wheel.

$$V = \pi(R_2^2 - R_1^2)h$$

(1)

- (ii) Beskou die wiel as opgebou uit dun ringe met dikte dR . Gee 'n uitdrukking vir die volume dV van 'n ring met straal R , dikte dR en hoogte h .

The cylinder can be constructed from a number of thin concentric cylindrical rings of thickness dR . Give an expression for the volume dV of a ring with radius R , thickness dR and height h .

$$dV = \pi R^2 dR h$$

(1)